

individual assignment 5

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# arranging plots
library(patchwork)

# coloring
library(wesanderson)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```
# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))
```

Cleaning

sites object

```
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

taxon_list object

```
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

water_quality object

```
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
```

```

unite("date_site", date, site, remove = TRUE) |>
# group by date_site column
group_by(date_site) |>
# calculate median pH, salinity, DO
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

```

aq_ins object

```

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
             names_to = "taxon",
             values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>

```

```

# calculate average abundance per liter (sum abundance divided by 7.5 liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm = TRUE),
  site_full = fct_rev(site_full))

```

Inserting the needed class code

Salinity visual

```

# base layer
salinity <- ggplot(data = aq_ins_clean,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
    y = med_sal)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
  shape = 21,
  width = 0.1,
  size = 2) +
# wrapping x-axis text

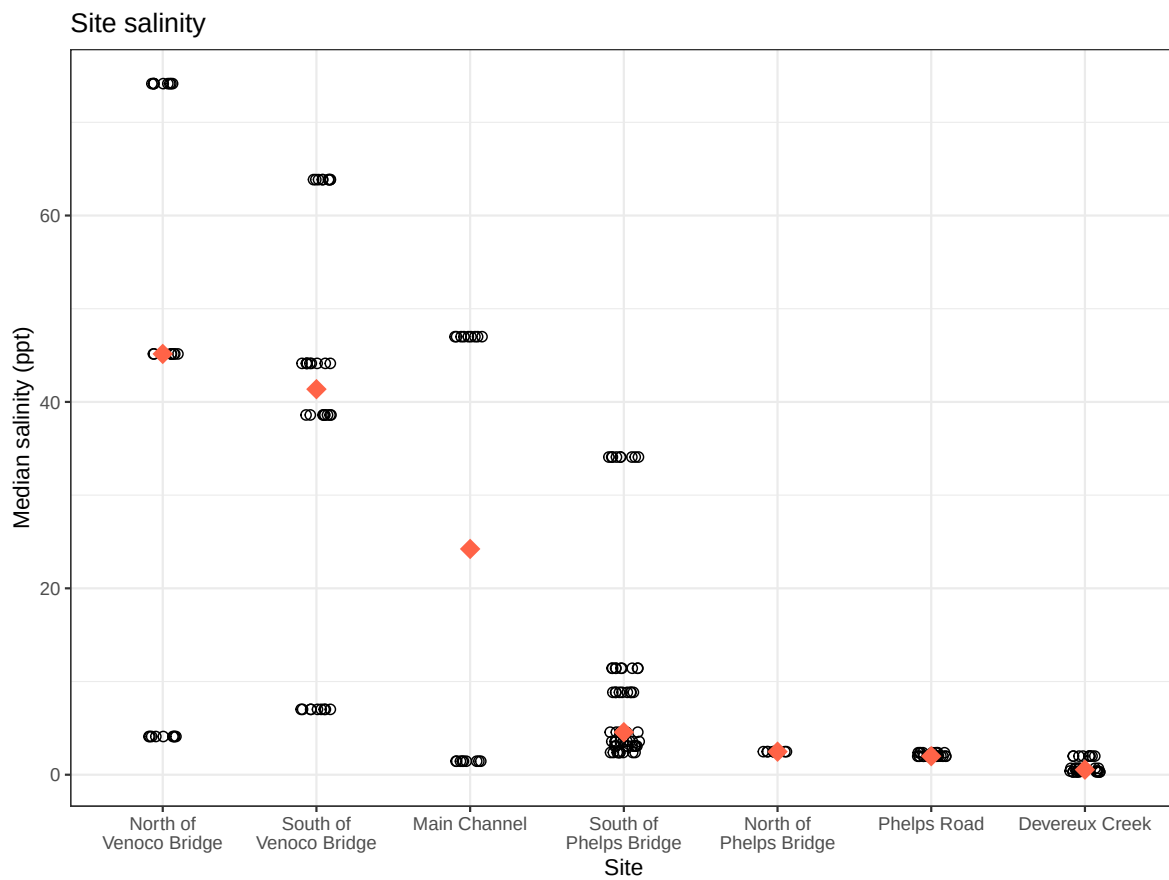
```

```

scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent median
stat_summary(fun = median,
             color = "tomato",
             size = 1,
             shape = 18) + # 18 = filled diamond
# relabelling x- and y-axes
labs(x = "Site",
     y = "Median salinity (ppt)",
     title = "Site salinity") +
# different theme
theme_bw()

# displaying object
salinity

```



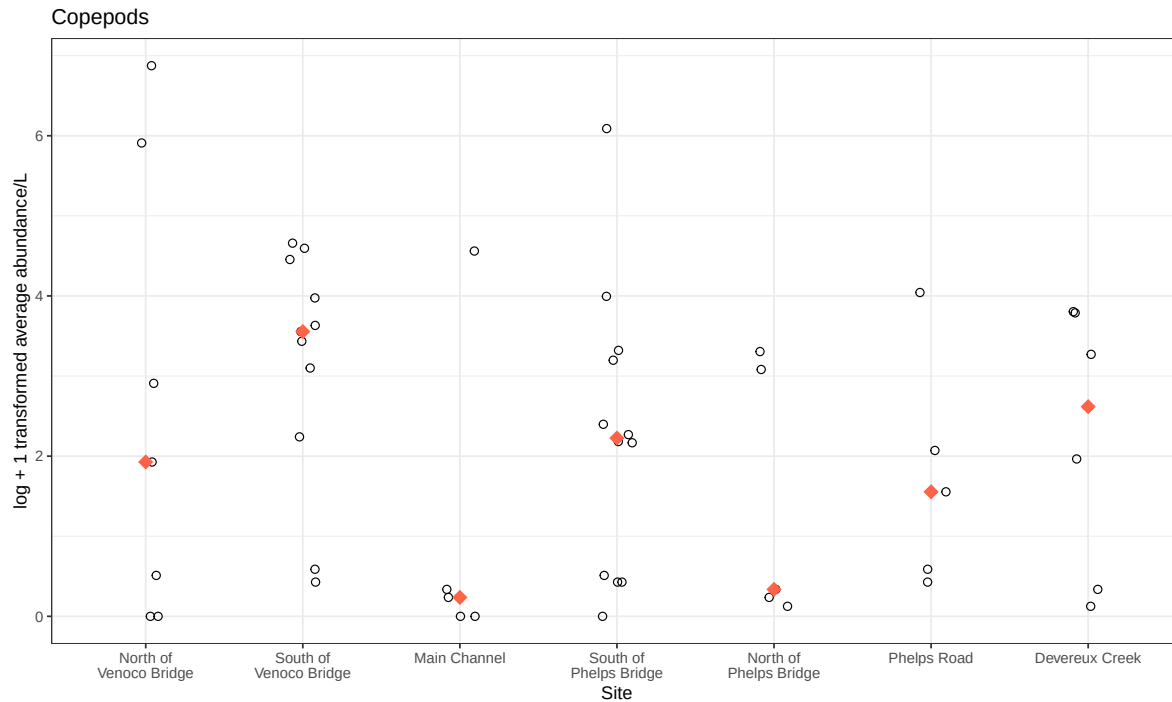
Copepod cleaning

```
# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

Copepod visual

```
# base layer
copepods <- ggplot(data = copepods,
  # x-axis: site
  # y-axis: mean abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
    shape = 21,
    width = 0.1,
    size = 2) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
    fun = "median",
    size = 4,
    color = "tomato",
    shape = 18) + # 18 = filled diamond
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
    y = "log + 1 transformed average abundance/L",
    title = "Copepods") +
  # different theme
  theme_bw()

# displaying object
copepods
```



Problems

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: site
- y-axis: average abundance/L
- geometry to represent average abundance/L: `geom_jitter()`
- geometry to represent medians: `stat_summary()`

b. Wrangle your data

```
# creating new object from clean aquatic inverts
ostracod <- aq_ins_clean |>
  # filter to only include ostracods
  filter(taxon == "ostracod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

```
# display 10 random rows
slice_sample(ostracod, n = 10)
```

```
# A tibble: 10 x 24
```

	date_site <chr>	date <dtm>	site <chr>	taxon <chr>	ave_lit <dbl>	med_ph <dbl>	med_sal <dbl>	med_do <dbl>
1	2021-11-16_NPB	2021-11-16 00:00:00	NPB	ostr~	72.1	7.3	2.38	12.1
2	2023-08-12_NEC	2023-08-12 00:00:00	NEC	ostr~	0	NA	NA	NA
3	2024-01-28_NPB	2024-01-28 00:00:00	NPB	ostr~	3.33	7.43	3.08	6.6
4	2022-11-12_NEC	2022-11-12 00:00:00	NEC	ostr~	0	7.33	7.01	2.4
5	2022-08-15_NMC	2022-08-15 00:00:00	NMC	ostr~	0.133	7.61	NA	NA
6	2023-02-12_NEC	2023-02-12 00:00:00	NEC	ostr~	2.13	NA	NA	NA
7	2023-01-28_NPB1	2023-01-28 00:00:00	NPB1	ostr~	0	7.66	NA	9.74
8	2023-11-12_NMC	2023-11-12 00:00:00	NMC	ostr~	0.533	NA	47	14.0
9	2023-06-06_NPB1	2023-06-06 00:00:00	NPB1	ostr~	0.533	7.48	NA	2.38
10	2023-05-04_NEC	2023-05-04 00:00:00	NEC	ostr~	0.667	8.05	NA	8.55

```
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

```
# creating the ostracod figure
ostracod_plot <- ggplot(data = ostracod,
  # x-axis: site
  # # y-axis: mean abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  # first layer: individual observations as open circles
  geom_jitter(height = 0,
    shape = 21,
    width = 0.1,
    size = 2) +
  # second layer: median as a filled blue diamond
  stat_summary(geom = "point",
    fun = "median",
    size = 4,
    color = "tomato",
```

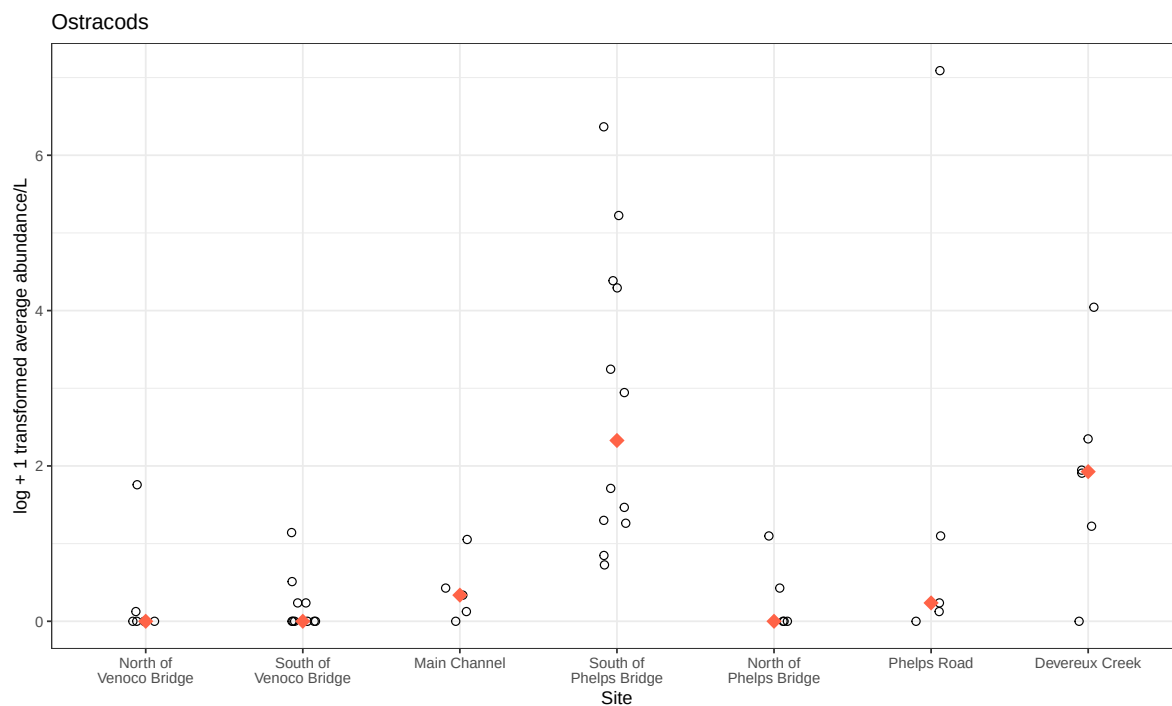


```

        shape = 18) + # 18 = filled diamond
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# axis labels and title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracods") +
# different theme
theme_bw()

# displaying object
ostracod_plot

```

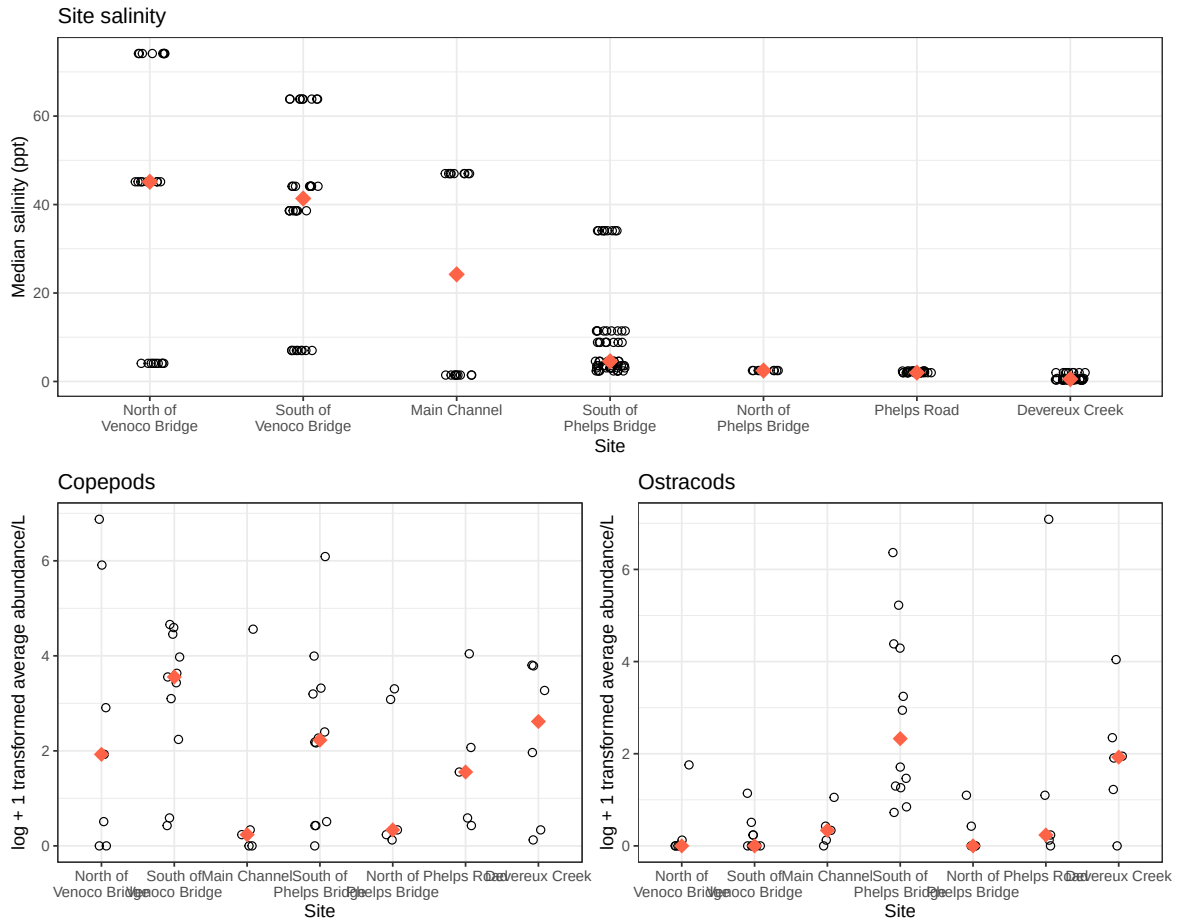


d. Create a multipanel figure

```

# combining the figures together
salinity/ ( copepods + ostracod_plot )

```



e. Interpret your figure

- Differences between sites in copepod and ostracod abundance:

Copepod abundance was highest at South of Venoco Bridge with a median above 3, while ostracod abundance peaked at South of Phelps Bridge with a median above 2. Overall, ostracods had lower median abundance than copepods at every site, suggesting copepods are the more dominant group across this system.

- How ostracod abundance may relate to salinity:

Ostracod abundance does not show a clear relationship with salinity — while most sites had low ostracod abundance regardless of their salinity level, South of Phelps Bridge had the highest ostracod abundance despite having low salinity (below 7 ppt), similar to several other low-salinity sites where ostracods remained scarce. This suggests that salinity alone does not

explain the variation in ostracod abundance across sites, and other environmental factors may be at play.

2. Visualizing total abundance as a function of salinity

a. Plan your figure

- x-axis: med_sal
- y-axis: log_ave_li
- color: taxon
- geometry: geom_point()
- facets: facet_wrap(~ taxon, scales = "free")

b. Wrangle your data

```
# modify the aq_ins_clean data frame
aq_ins_clean <- aq_ins_clean |>
  # convert taxon to title case
  mutate(log_ave_lit = log(ave_lit + 1), # log + 1 transform average abundance per liter
         taxon = str_to_sentence(taxon), # sentence case - only first letter capitalized
         taxon = str_replace_all(taxon, "_", " "), # replace underscores with spaces
         taxon = fct_reorder(taxon, ave_lit, .fun = max)) # reorder taxon factor levels by maximum abundance

# display 10 random rows
slice_sample(aq_ins_clean, n = 10)
```

A tibble: 10 x 24

	date_site <chr>	date <dtm>	site <chr>	taxon <fct>	ave_lit <dbl>	med_ph <dbl>	med_sal <dbl>	med_do <dbl>
1	2022-05-11_NDC	2022-05-11 00:00:00	NDC	Ostr~	0	9.11	0.398	9.4
2	2022-08-10_NPB	2022-08-10 00:00:00	NPB	Anne~	0	8.28	8.84	9.81
3	2022-05-06_NMC	2022-05-06 00:00:00	NMC	Ostr~	0	9.96	1.45	16.5
4	2022-11-19_NPB1	2022-11-19 00:00:00	NPB1	Anne~	0	7.26	2.47	3.96
5	2022-08-15_NMC	2022-08-15 00:00:00	NMC	Cope~	0	7.61	NA	NA
6	2022-02-23_NVBR	2022-02-23 00:00:00	NVBR	Hemi~	0	8.86	45.2	10.2
7	2023-06-06_NPB1	2023-06-06 00:00:00	NPB1	Anne~	0.267	7.48	NA	2.38
8	2022-02-23_NPB1	2022-02-23 00:00:00	NPB1	Anne~	0	NA	NA	NA
9	2022-08-08_NPB2	2022-08-08 00:00:00	NPB2	Ostr~	2	7.68	2.36	2.43
10	2021-09-01_NPB2	2021-09-01 00:00:00	NPB2	Clad~	0.667	NA	NA	NA

i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,

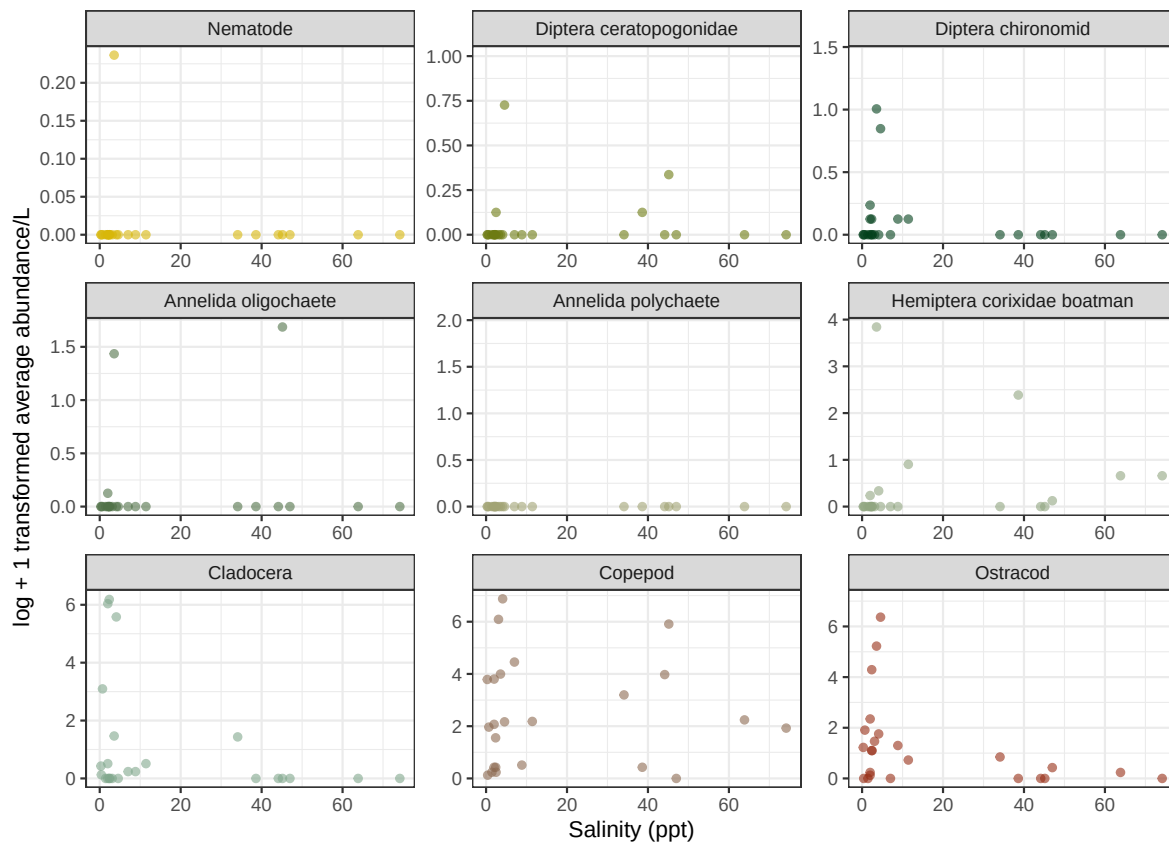
```
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

```
# start with aq_ins_clean data frame
ggplot(data = aq_ins_clean,
       mapping = aes(x = med_sal, # x-axis: salinity in ppt
                     y = log_ave_lit, # y-axis: log + 1 transformed average abundance/L
                     color = taxon)) + # color by the taxon
# points to show relationship between salinity and abundance
geom_point(alpha = 0.6) +
# separate panel for each taxon with free scales on both axes
facet_wrap(~ taxon, scales = "free") +
# readable axis labels
labs(x = "Salinity (ppt)",
     y = "log + 1 transformed average abundance/L",
     title = "Invertebrate Abundance as a Function of Salinity") +
# different theme
theme_bw() +
# different colors from default
scale_color_manual(values = wes_palette("Cavalcanti1",
                                         n = n_distinct(aq_ins_clean$taxon),
                                         type = "continuous")) +

# remove legend
theme(legend.position = "none")
```

Invertebrate Abundance as a Function of Salinity



d. Interpret your figure

Copepods and ostracods both displayed a tall, dense vertical scatter of points concentrated on the left side of the graph (salinity 0–10), indicating peak abundance at low salinities, though copepods uniquely showed a secondary cluster of high abundance points at approximately salinity 40, distinguishing them from all other taxa. In contrast, nematodes, annelid oligochaetes, cladocerans, diptera chironomids, and annelid polychaetes all showed a flat, compressed scatter pattern hugging the x-axis across the entire salinity range, suggesting consistently low abundance regardless of salinity. The vertical spread of copepod and ostracod points at low salinity was notably taller than any scatter observed in the remaining taxa, indicating their maximum abundances were substantially higher overall. Ostracods and the low-abundance taxa were similar in showing no rebound at higher salinities, whereas copepods were the only taxon whose scatter pattern suggested a second peak.